

Learned Epistemic Interrupts: Main-Model Control versus Semantic and Hidden-State Sidecars

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Abstract

Self-RAG already establishes learned on-demand retrieval and reflection in the main generative LM [2]; learning when to retrieve is therefore not our novelty claim. We instead test where epistemic-control competence should live: inside the main LM, in a model-independent text sidecar observing ordinary generation prefixes, or in a lightweight probe over a frozen LM’s hidden state. Controllers must choose among NONE, DB, LEXICAL, VECTOR, and WEB rather than only retrieve/no-retrieve. We compare quality, unsupported claims, routing, transfer, and cost without assuming that any architecture is superior.

1 Problem Decomposition

Separate:

$$P(I_t \mid h_t)$$

for whether retrieval is needed,

$$P(S_t \mid I_t, h_t)$$

for source selection, and

$$P(Q_t \mid S_t, I_t, h_t)$$

for normalized query formation. Detection, routing, and query formation must be evaluated independently.

2 Candidate Interfaces

Compare:

1. FLARE-like confidence control [1];
2. Self-RAG-style main-model learned retrieval/reflection control;
3. a text sidecar performing DETECT/ROUTE/NORMALIZE over ordinary prefixes;
4. a matched lightweight hidden-state sidecar with the main LM frozen;
5. oracle trigger, source, and query.

If exact Self-RAG training is infeasible, the condition must be called a Self-RAG-inspired approximation rather than Self-RAG proper. Main-model training is confined to that declared comparison; text and hidden-state sidecars keep the generative LM frozen.

3 Training Data

Use realistic generation prefixes, not only original user questions. Label:

- retrieve / do-not-retrieve;
- information-need type;
- preferred source;
- canonical query/IR;
- freshness/urgency;
- evidence availability;
- epistemic role.
- miss-risk;
- whether the need is computational rather than epistemic.

Hard negatives must include facts already present in context, quotations, hypothetical or counterfactual statements, questions, user-given facts, computational needs better served by calculators or logic, and fluent non-factual prose.

4 Epistemic Roles

Distinguish imminent factual commitment, verification request, uncertainty, quotation, hypothesis, counterfactual, user-provided fact, and computed or derived fact. This prevents brittle surface matching from turning every factual-looking phrase into a hard retrieval event.

5 Routing

Predict among

{NONE, DB, LEXICAL, VECTOR, WEB},

with optional KG. Support abstention and, when justified, a small top- k source set. The DB, lexical, vector, and web actions must retain the distinct query and execution semantics established in Paper 2.5.

6 Sidecar Hypotheses and Ablations

Model independence, upgradeability, routing specialization, small-model leverage, and lower re-training cost are hypotheses rather than premises. Compare confidence-only, semantic-only, and semantic-plus-confidence sidecars at matched complexity and runtime cost. The hidden-state probe must also be complexity- and cost-matched to the text controller; it earns no native-interface claim merely by accessing activations.

For cross-model transfer, train the text sidecar on traces from one or several model families and evaluate it on a different frozen family. Repeat across model sizes. Successful transfer supports reusable runtime control; architecture- or task-dependent winners are equally valid outcomes.

7 Calibration and Cost

False retrievals cost latency/money; missed retrievals cost unsupported claims. Evaluate thresholded policies under

$$C = \lambda_{fp}C_{fp} + \lambda_{fn}C_{fn} + \lambda_{lat}C_{lat} + \lambda_{tok}C_{tok}.$$

Do not optimize only F1.

8 Experiments

Evaluate in-domain detection, paraphrase robustness, unseen entities/relations, domain and cross-model transfer, changing freshness windows, multi-factual answers, and model-size effects. Measure the high-confidence/high-risk and low-confidence/low-risk quadrants explicitly. Include DETECT/ROUTE/NORMALIZE, confidence-feature, and action-space ablations.

9 Baselines

1. no retrieval;
2. upfront RAG;
3. explicit search/tool calls;
4. Paper 2.5 heuristic reflex routing;
5. FLARE-like confidence controller;
6. Self-RAG-style main-model learned controller;
7. text sidecar DETECT/ROUTE/NORMALIZE;
8. frozen-LM hidden-state sidecar;
9. oracle trigger/source/query.

10 Metrics

Report trigger AUROC/AUPRC and P/R, high-confidence/high-risk recall, low-confidence/low-risk false retrieval, calibration overall and by source, selective-risk/coverage curves, routing top-1/top- k and confusion matrices, normalization correctness, unsupported claims, final accuracy, count/fanout, latency/cost, model-size interactions, OOD and cross-model degradation, and the cost versus unsupported-claim Pareto frontier.

11 Falsification

The sidecar architecture is weakened if main-model Self-RAG-style control dominates quality and cost, sidecars do not transfer, heterogeneous routing adds no value, confidence explains semantic gains, hidden-state dependence is too model-specific, or the detector memorizes surface phrases. A main-model, sidecar, hybrid, or task/model-size-dependent winner is a valid result.

12 Scope

No joint base-model training except the declared main-model comparison; no transactional or backtracking architecture as the main contribution; and no PRA or native KV dependency. Relevance and support labels are prior art, not a novelty claim. Paper 4 concerns persistent cross-source provenance, dependency propagation, transactional scopes, retraction, and backtracking.

References

- [1] Z. Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023. <https://arxiv.org/abs/2305.06983>.
- [2] A. Asai et al. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *ICLR*, 2024. <https://arxiv.org/abs/2310.11511>.